Ling 285
Human Language and Technology
Lecture 13

Masses and Springs in Speech Production

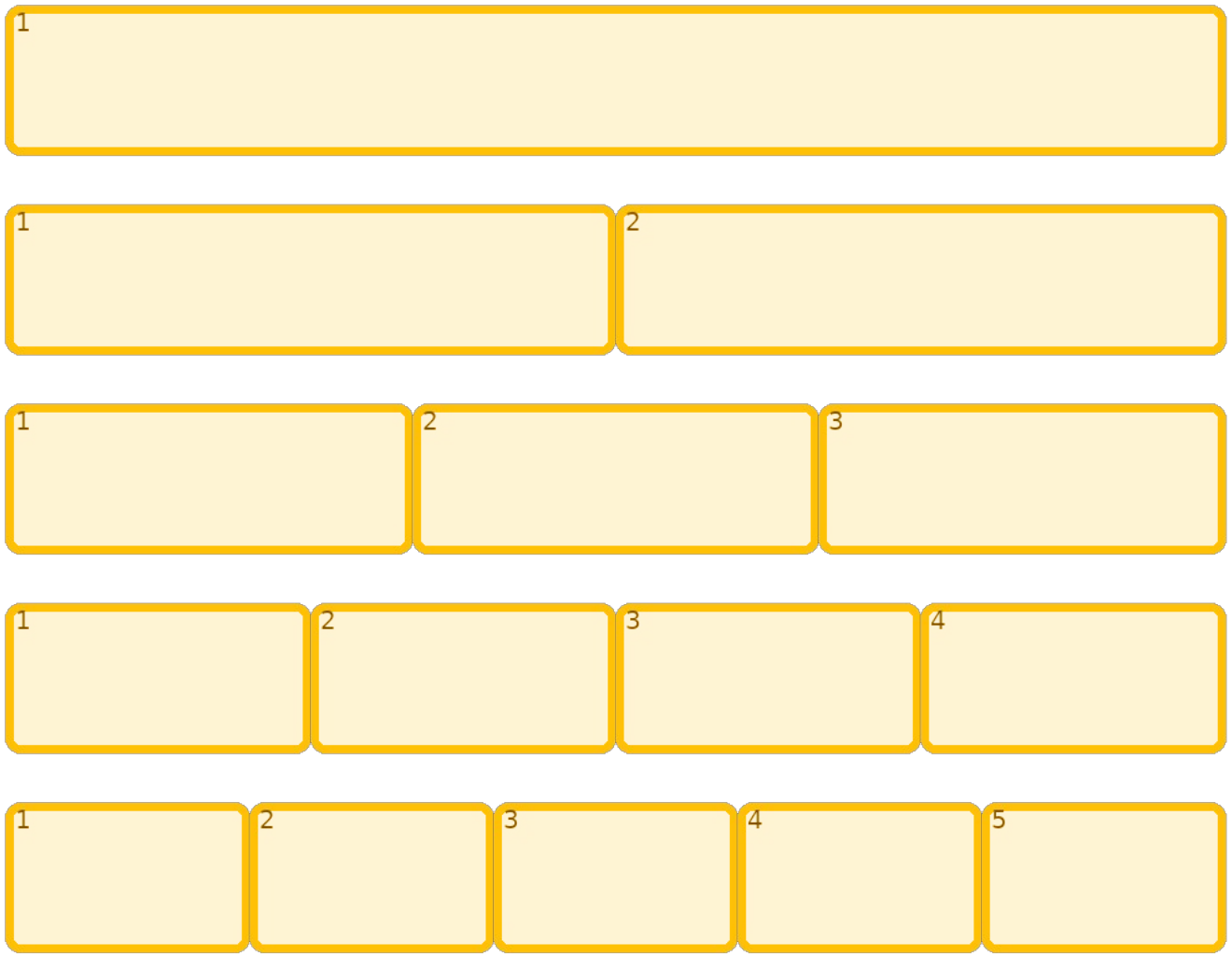
Khalil Iskarous

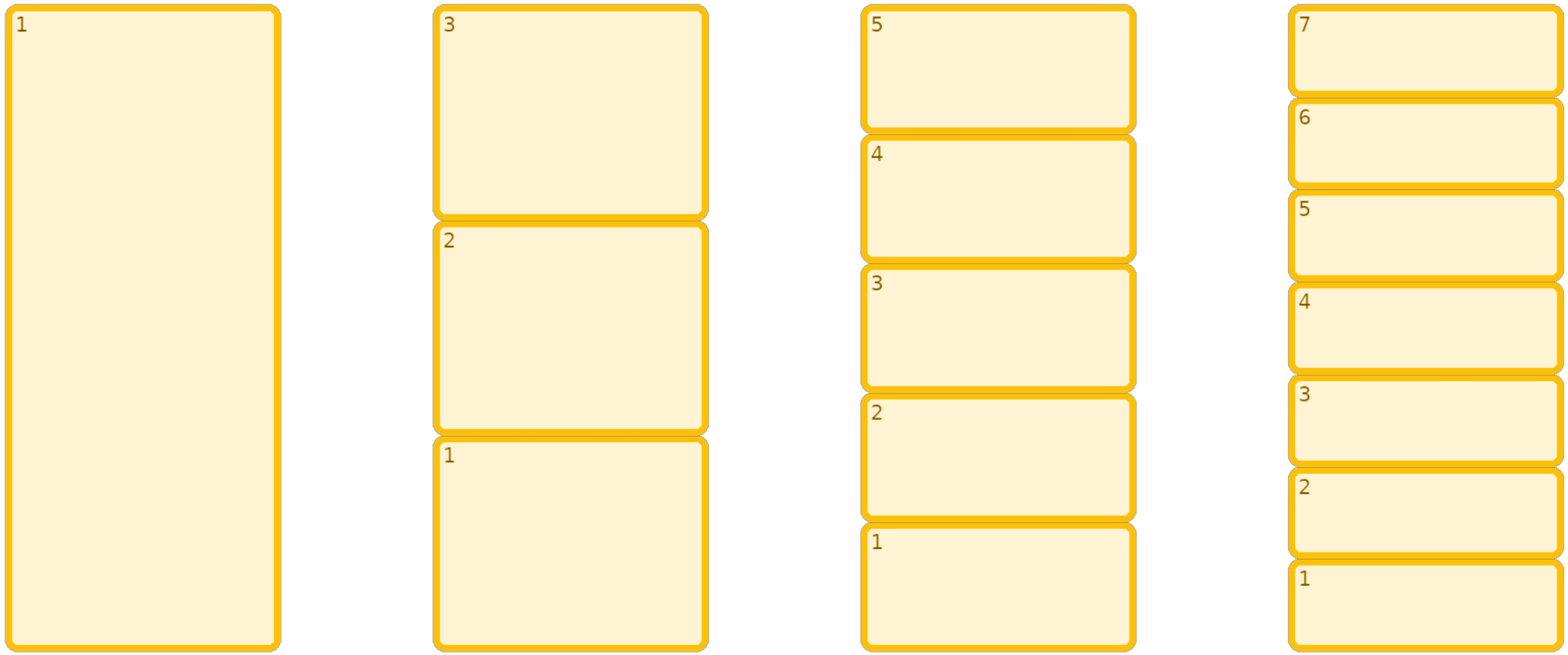
Department of Linguistics

University of Southern California

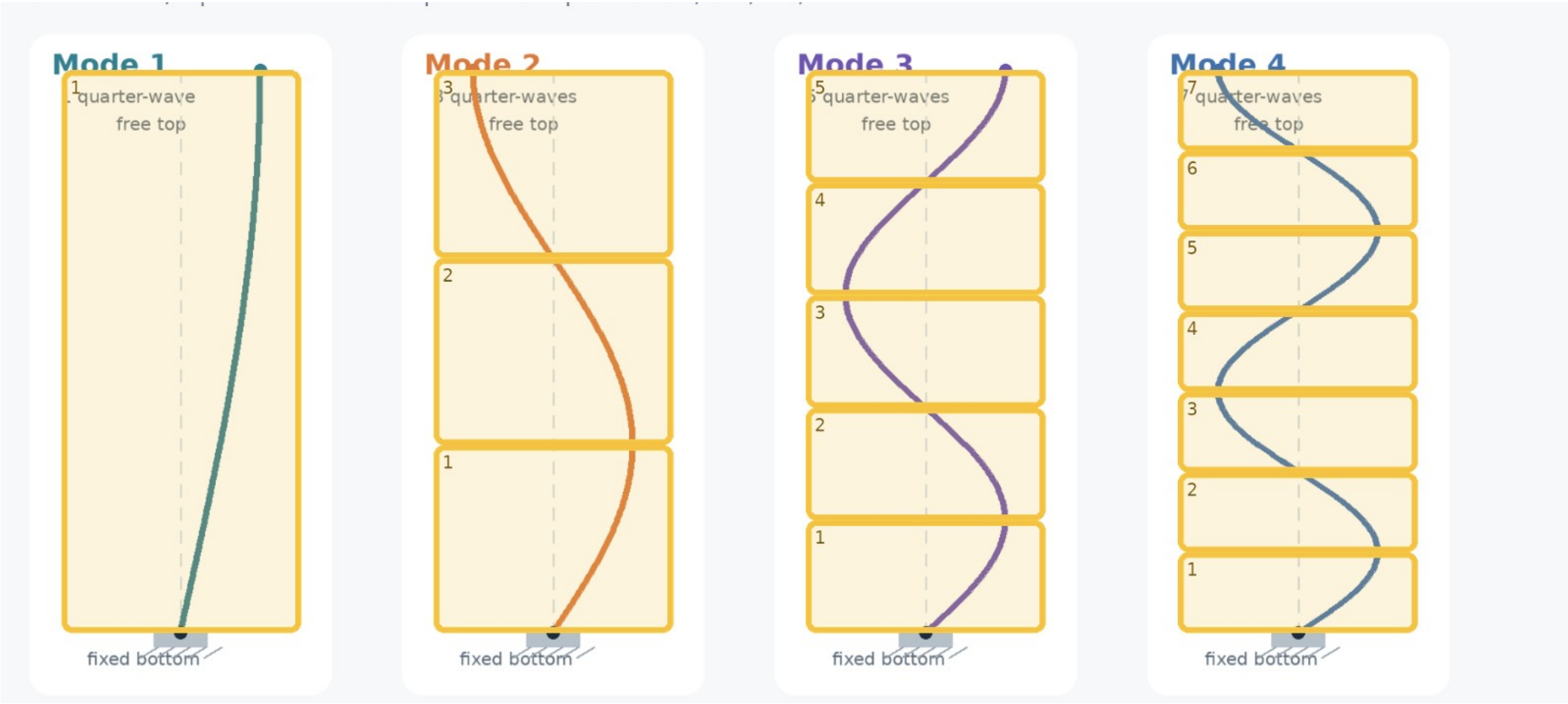
Principles of Mass-Spring Systems

1. Mass ↑ Frequency ↓, Stiffness ↑ Frequency ↑
2. An n-mass system has n modes of vibration, each with a higher frequency
3. An m's increase lowers frequency proportionally to how much the mass moves, which depends on where it is!
4. A k's increase raises frequency proportionally to how much the spring stretches/compresses, which depends on where it is!
5. Where there's maximal movement, there's no stretching, An where there's maximal stretching, there's no movement!
6. In a Fixed-Fixed System, the *harmonics* are multiples of the base frequency, because the base mode fits n-times into each successive mode
7. In a Fixed-Free System, the *formants* are odd multiples of the base frequency, because the base mode fits an odd n-times into each successive mode


1
1
2
1
2
3
1
2
3
4
1
2
3
4
5


1
3
2
1
5
4
3
2
1
7
6
5
4
3
2
1

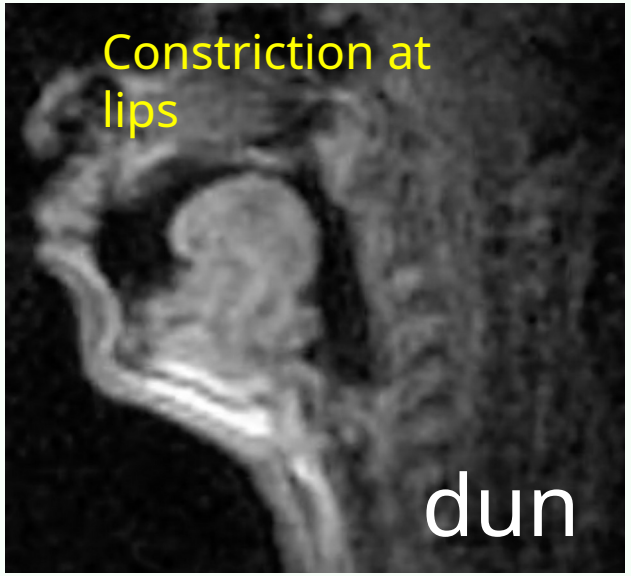
Pop Quiz


Mode 1
1
quarter-wave
free top
fixed bottom
Mode 2
3
quarter-waves
free top
2
1
fixed bottom
Mode 3
5
quarter-waves
free top
4
3
2
1
fixed bottom
Mode 4
7
quarter-waves
free top
6
5
4
3
2
1
fixed bottom

Vowels

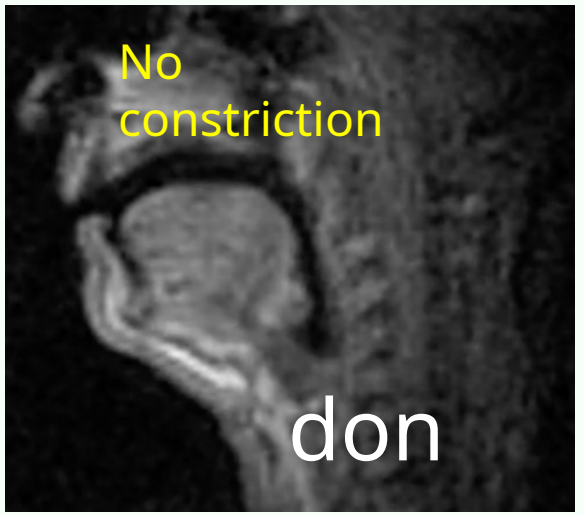
High





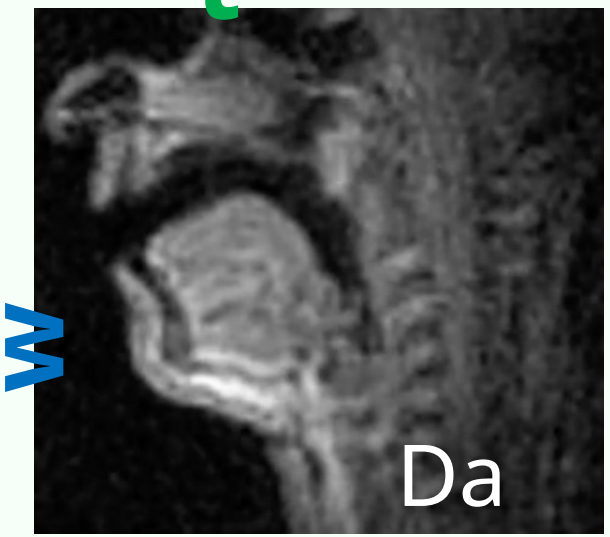
Front

Back

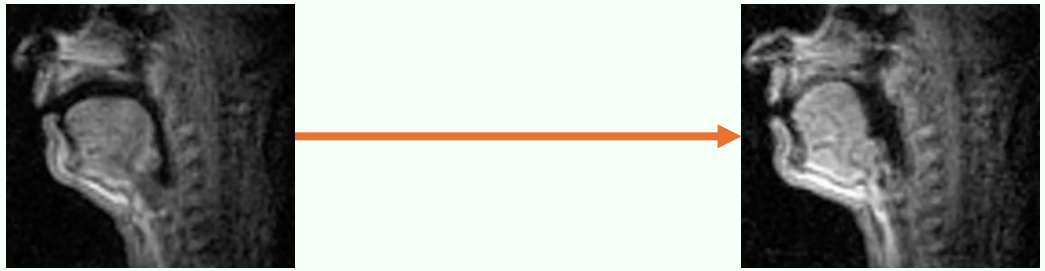


Central

Low







When we compress air, as happens in a constriction, its stiffness rises, so making a constriction at a place that experiences lots of compression, will raise frequency. And bunching up air makes its mas//density larger, so in a part of the vocal tract where air moves a lot, constricting raises mass! So making a constriction increases mass and increases stiffness.

But be careful!!!! This doesn’t mean that constrictions have no effect on frequency!!! Why? Because the parts of a vibrating medium that moves the most alternate with those that compress/uncompress the most!

/u/ = free-end loading

Mode 1: motion dominates

Mode 2: motion dominates

F1 ↓ F2 ↓

Click to reveal the reasoning steps.

/i/ = constriction near 2/3

Mode 1: mass motion dominates

Mode 2: springiness dominates

F1 ↓ F2 ↑

Click to reveal the reasoning steps.

/a/ = constriction near 1/3

Mode 1: springiness dominates

Mode 2: mass motion dominates

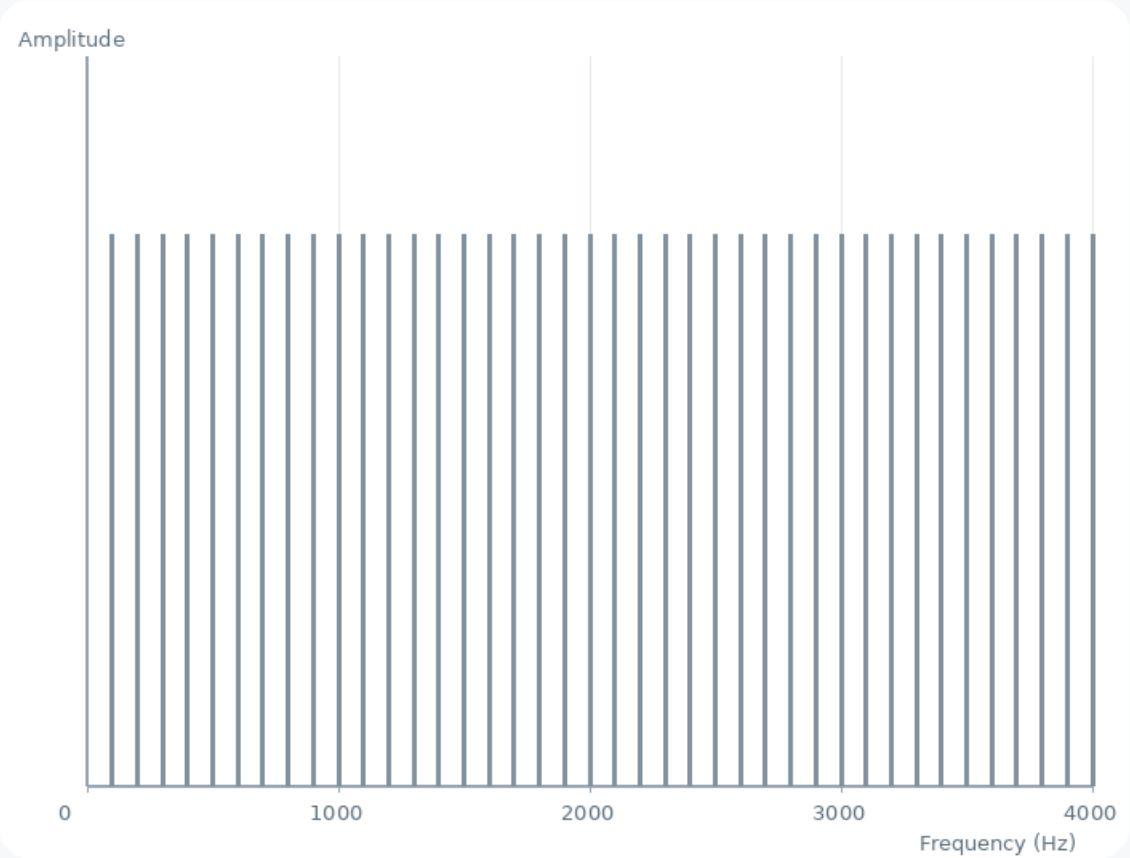
F1 ↑ F2 ↓

Click to reveal the reasoning steps.

Source–filter theory: /ə/ schwa

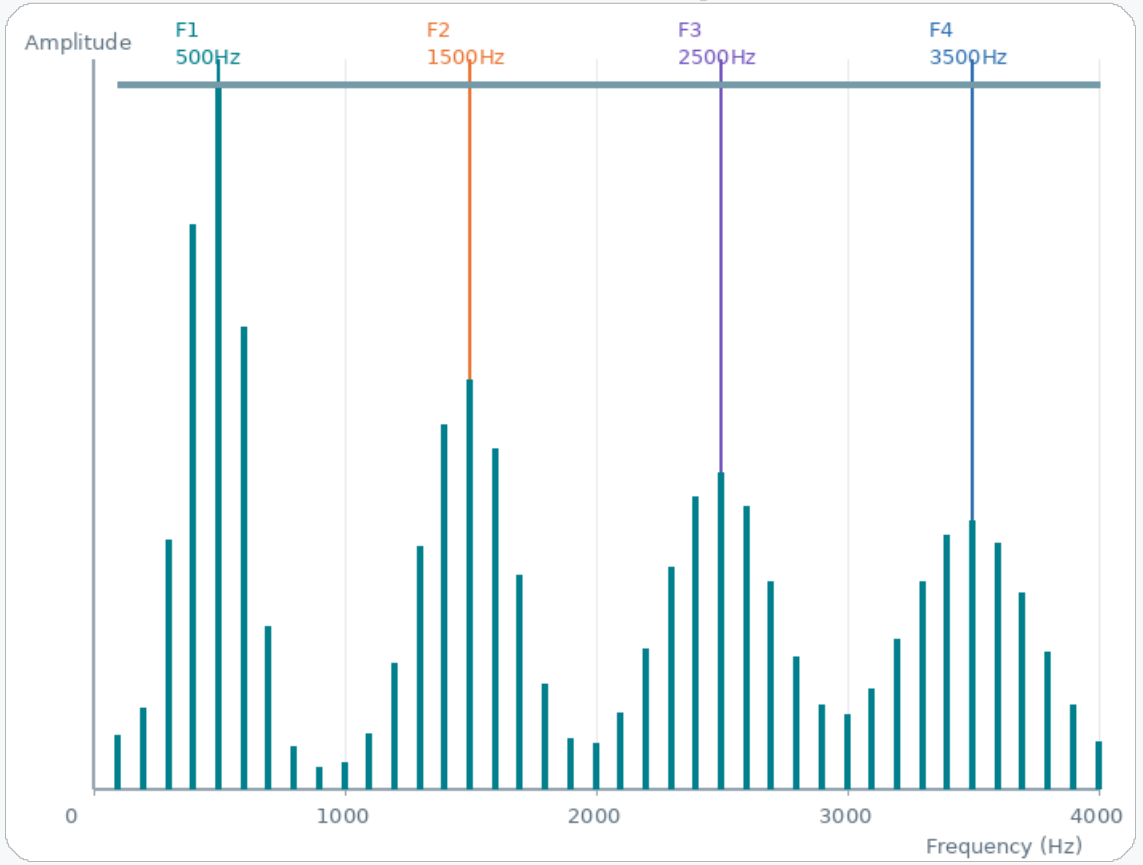
The glottal source gives a harmonic comb. The vocal tract filter sculpts the heights of those harmonics.

Source: harmonics every 100 Hz



Before the vocal tract: evenly spaced vertical lines

Filter + source: formants sculpt /ə/ schwa

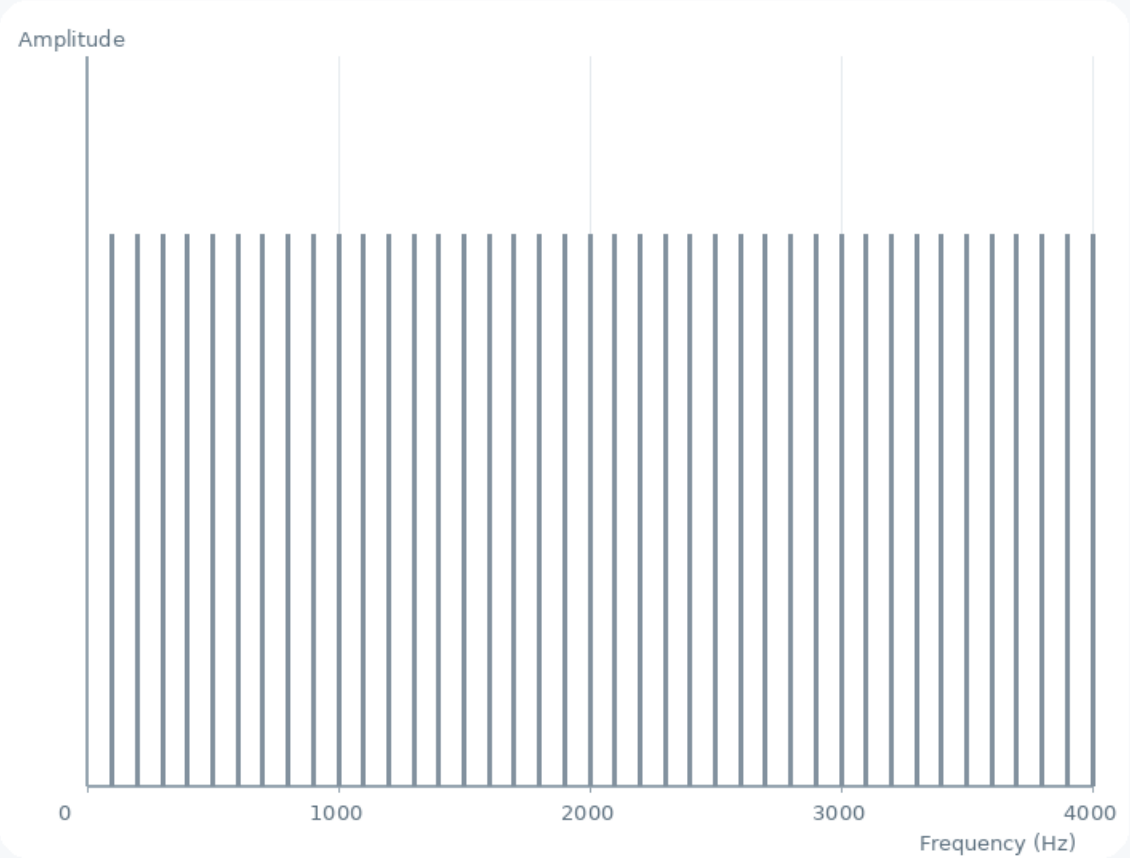


Same harmonics, different heights: the envelope is the vowel

Source–filter theory: /i/

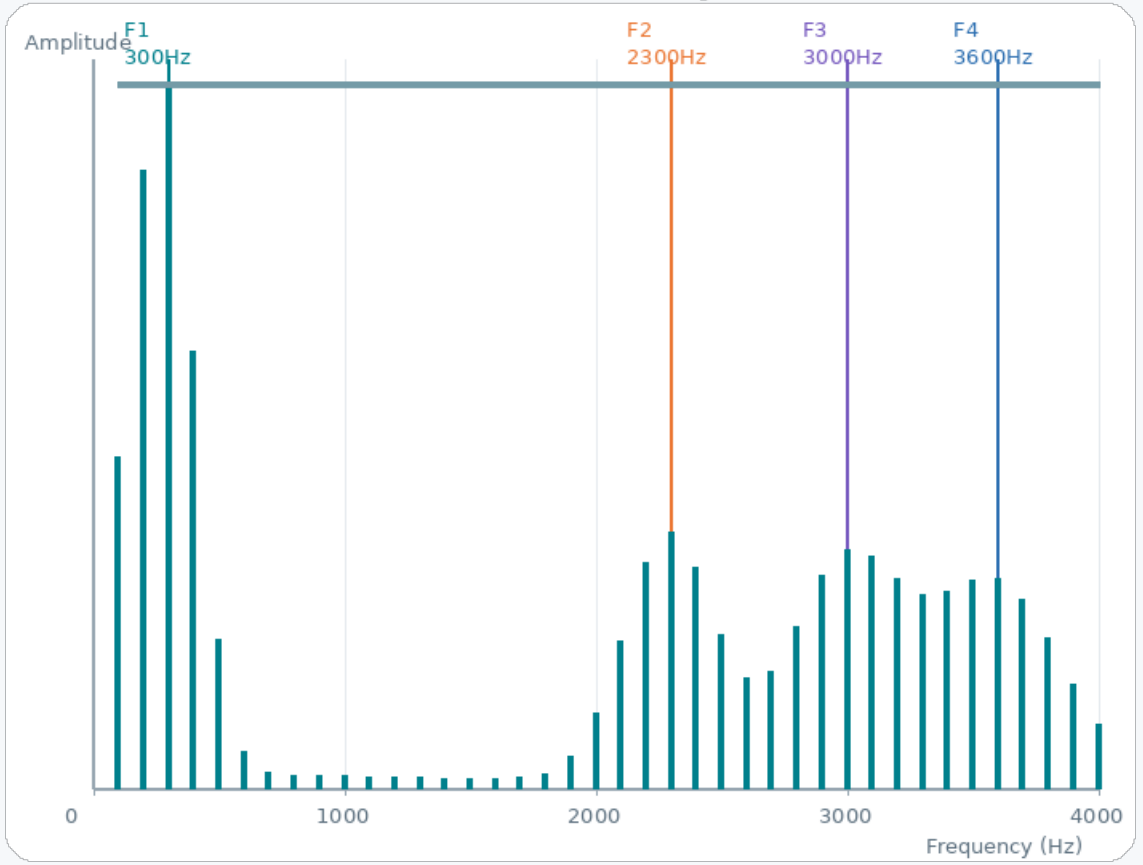
The glottal source gives a harmonic comb. The vocal tract filter sculpts the heights of those harmonics.

Source: harmonics every 100 Hz



Before the vocal tract: evenly spaced vertical lines

Filter + source: formants sculpt /i/

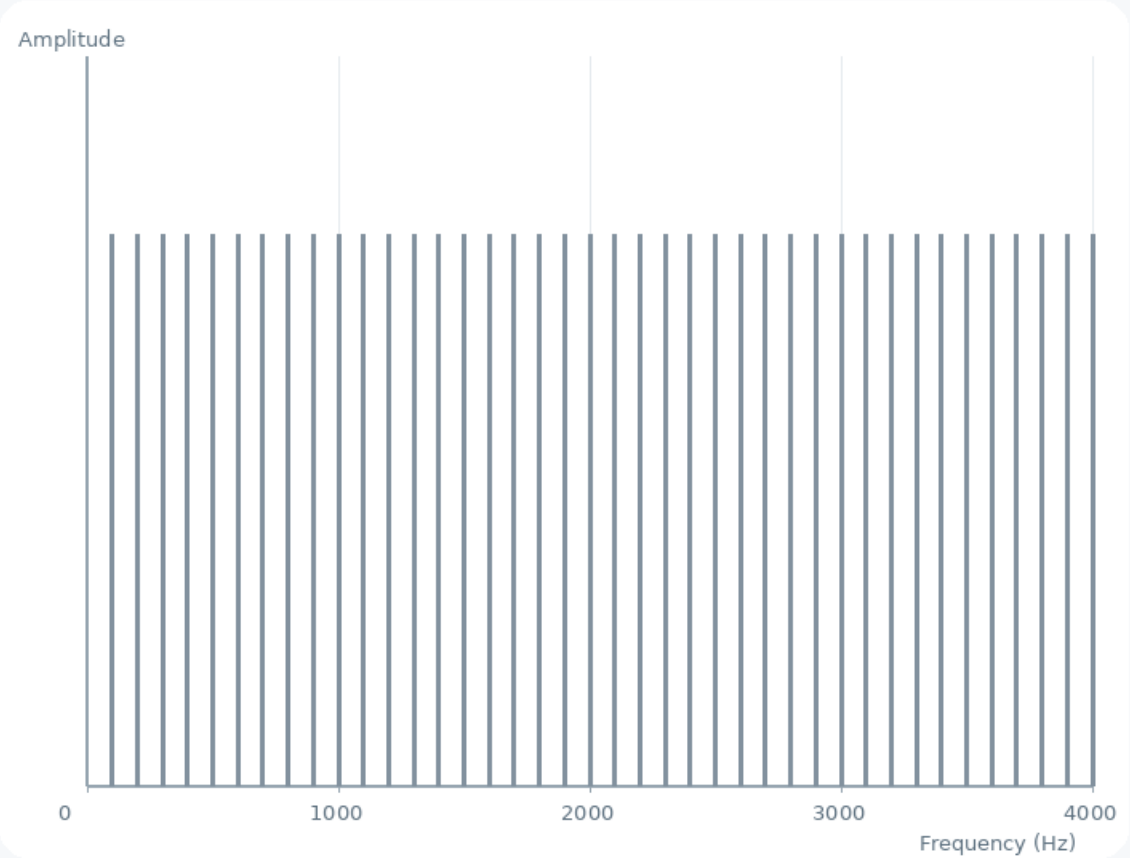


Same harmonics, different heights: the envelope is the vowel

Source–filter theory: /a/

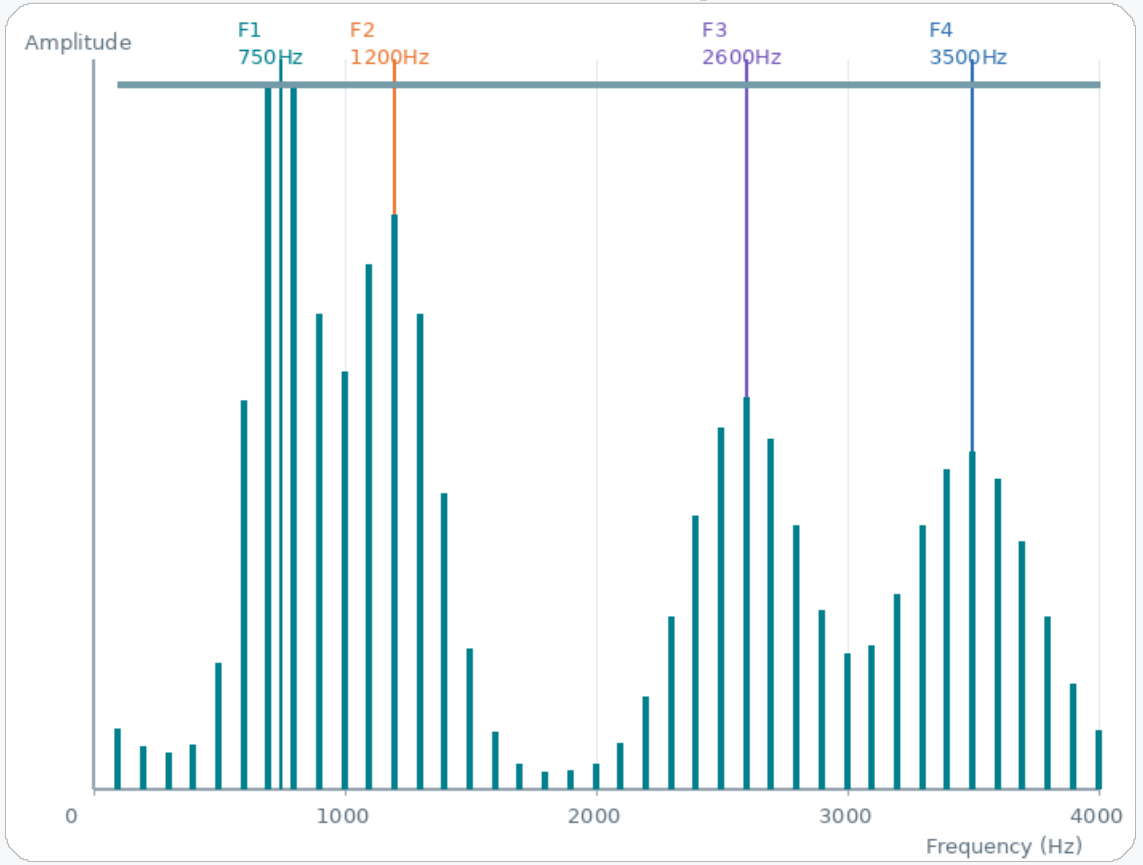
The glottal source gives a harmonic comb. The vocal tract filter sculpts the heights of those harmonics.

Source: harmonics every 100 Hz



Before the vocal tract: evenly spaced vertical lines

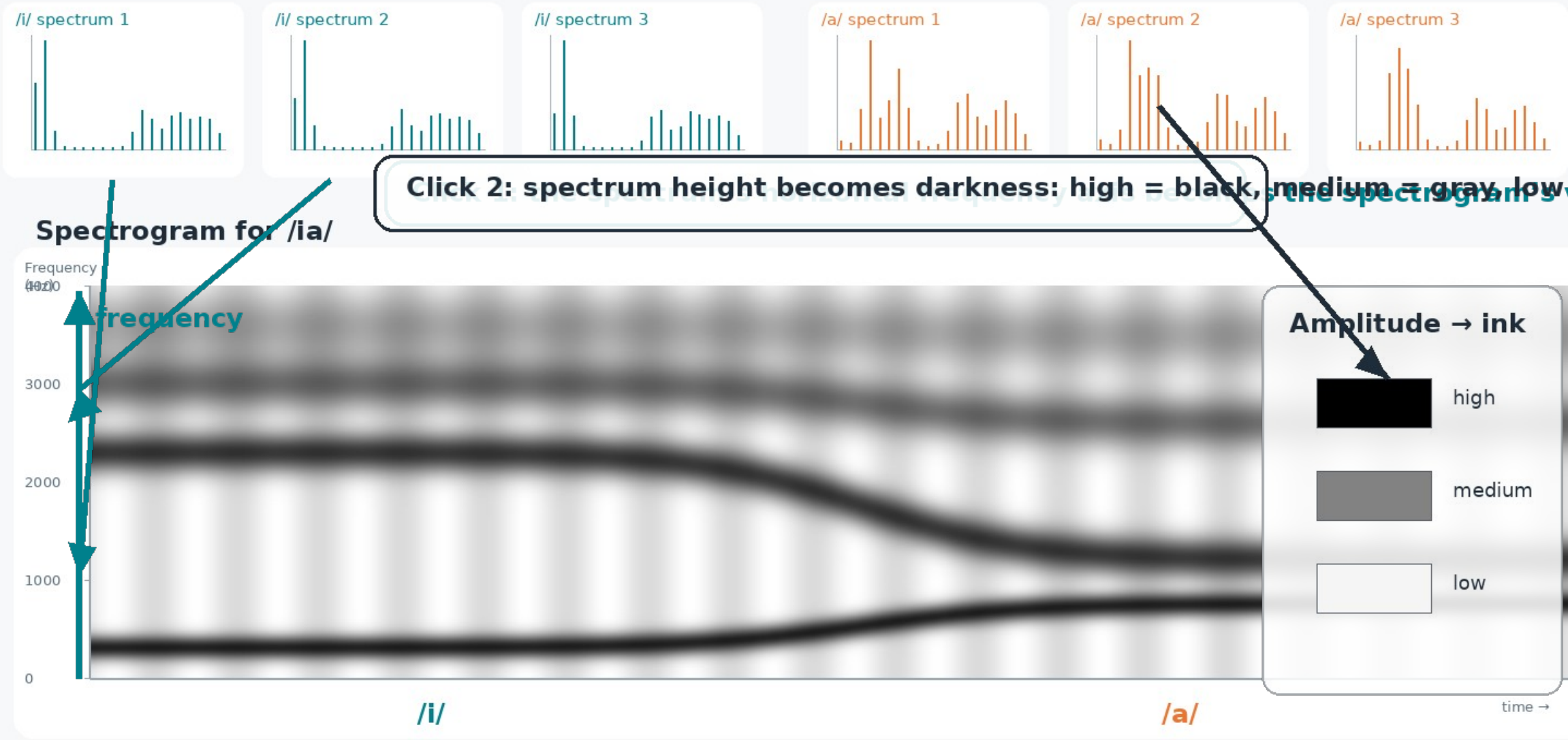
Filter + source: formants sculpt /a/



Same harmonics, different heights: the envelope is the vowel

Spectrum versus spectrogram

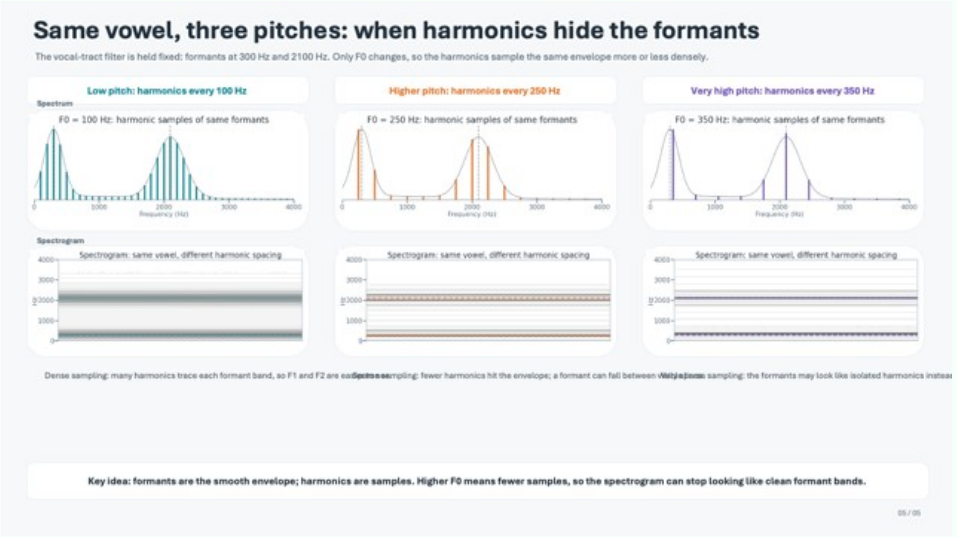
A spectrum is one instant. A spectrogram stacks many spectra over time.



Where were you a year ago?

Problem: Harmonics and Formants

- Imagine you have harmonics 100,200,300,400, etc., and F1 at 300, and F2 at 2100. In that case, the formants can structure the amplitudes of many harmonics
- Vs. harmonics 250, 500, 750…. ☹
- It's very hard to tell harmonics from formants in the latter case :(
- Consider: a toddler's harmonics could be 400,800…
- A soprano: 800,1600, 2400…



HW4

- You will figure out the formants for *your* basic vowels in Lab on Monday, and you will be given credit for HW4 by your TA.